

Heavy metal adsorption from aqueous solution using *Eichhornia crassipes* dead biomass

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ABSTRACT

The adsorption of Zn(II), Cd(II) and Cr(VI) from single, bi- and tri-metal systems was studied with respect to varying metal and biomass concentration, pH and agitation time on *Eichhornia crassipes* dead biomass. The maximum adsorption of Cd(II) and Zn(II) was observed at pH 5 and 6 respectively, whereas that of Cr at pH 2.0. An enhanced metal adsorption on 0.5 g of biomass with an agitation time of 120 min for Zn(II) and Cd(VI) and 180 min for Cr was recorded. Maximum adsorption of Cd(II) (12.4 mg g^{-1}), obtained from initial concentration (40 mg l^{-1}), followed by Zn(II) (9.3 mg g^{-1}) and Cr (5.6 mg g^{-1}) at optimal conditions. Cd adsorption capacity increased in the bi-metal system whereas that of Zn and Cr reduced in both bi- and tri-metal systems. After elution, higher metal recovery was recorded from a single metal system than a trimetal system. *E. crassipes* dead biomass retained the metal adsorption capacity up to two cycles, indicating its prospects for the wastewater treatment and metal recovery.

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1. Introduction

Application of microbial or plant biomass could be an effective, low cost and ecofriendly means for removal and/or recovery of toxic metals from contaminated wastewater and industrial effluents (Figueira and Ribeiro, 2005; Pal et al., 2009) even at low concentrations. Certain aquatic and terrestrial plants, composed of lignin (Garcia-Valls and Hatton, 2003), cellulose (Shukla and Sakhardande, 1991), hemi-cellulose, pectins (Nawirska, 2005), phytic acid (Martin and Evans, 1987) and many proteins (Mejare and Bulow, 2001), offer active sites such as carbonyl (C=O), carboxyl (–COO), hydroxyl (–OH), amino (–NH₂) and sulfhydryl (–SH) groups for binding of metal cations (Vaughan et al., 2001; Shin and Rowell, 2005), and make them popular for removing metals from contaminated waters. Although, high potential of live aquatic weeds viz. *Pistia stratiotes*, *Spirodela intermedia*, *Lemna minor* (Miretzky et al., 2004) *Eichhornia crassipes* and *Valisneria spiralis* (Singhal and Rai, 2003; Verma et al., 2005) for removing toxic metals from the aqueous system have been suggested, the application of dead biomass is preferred, as it eliminates the problem of metal toxicity and the economic aspects of nutrient supply and growth maintenance (Sekhar et al., 2003; Verma et al., 2008). In this context, the dead biomass of *Salvinia natans* having a high adsorption capacity for Ni, Co, Cr, Fe, and Cd (Dhir et al., 2010), *Pinus ponderosa* bark for Cu, Zn, Cd and Ni (Oh and Tshabalala, 2007), *Azadirachta indica* (Singh et al., 2008) and *Parthenium hysterophorus*

(Venkateswarlu et al., 2007) for Cr(VI) from aqueous systems have been reported. To explore the adsorption potential, the present study was undertaken to optimize the conditions viz. pH, metal and biomass concentration and agitation time for removal of Zn(II), Cr(VI) and Cd (II) from single, bi- and tri-metal systems and also their recovery employing dead biomass of *E. crassipes*.

2. Material and methods

2.1. Biomass and metal solution preparation

E. crassipes is a fast growing, free-floating, perennial aquatic plant known for extremely high tolerance and uptake of heavy metals (Maine et al., 2006; Upadhyay and Tripathi, 2007; Abou-Shanab et al., 2007; Skinner et al., 2007). The plant has high cellulose content having carboxylate (–COO[–]) and hydroxyl (OH[–]) groups predominantly in the cell wall (Kelley et al., 1999). Live plants of *E. crassipes* collected from a pond at Karula Nala (Moradabad, India), washed with tap water followed by distilled water to eliminate the sediments, oven dried at 70 °C for 48 h, ground and sieved through 2 mm sieve. The 100 g biomass was pretreated with 1000 ml of 0.1 M NaOH for 30 min and washed thrice with deionized water till the pH of the wash solution reached neutral (Xinjiang, 2006).

Stock solutions (1000 mg l^{-1}) of Zn(II), Cd(II) and Cr(VI) were prepared by dissolving Zn(NO₃)₂, Cd(NO₃)₂ and K₂Cr₂O₇ respectively in deionized water. A single metal system having different metal concentrations (i.e. 10, 20, 30, 40 and 50 mg l^{–1}) was prepared by dilution with deionized water. pH of each metal system was adjusted by adding 0.01 M HNO₃ and NaOH. To determine adsorption behaviour of Zn(II), Cd(II) and

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Cr(VI) ions in bi- and tri-metal systems, the initial concentration of each of the metal ions was taken 20 mg l^{-1} and 40 mg l^{-1} separately at pH 5 ± 0.2 .

2.2. Equilibrium adsorption experiments

Metal adsorption experiment was performed in triplicate in batch manner by adding 0.5 g of biomass in 200 ml aliquots of Zn(II), Cd(II) and Cr(VI) solutions of varying initial concentrations (i.e. 10, 20, 30, 40 and 50 mg l^{-1}) at pH 5 and equilibrated at $25 \pm 2^\circ \text{C}$ by agitating in a rotary shaker at 150 rpm for 2 h. A control was also set up with no biomass addition. At the end of experiment, equilibrium concentration, C_e of the metal ion in the filtrate was analyzed by atomic absorption spectrophotometer (AAS, GBC AvantaVer. 1.33, Australia).

Adsorption behaviour of each metal ion was also studied in triplicate at varying pH (i.e. 2 to 7), biomass concentration (i.e. 0.2, 0.3, 0.4, 0.5, 0.6 and 0.7 g) and agitation time (i.e. 15, 30, 60, 120, 180 and 240 min) in a single metal system (40 mg l^{-1} , 200 ml) with 0.5 g of biosorbent at pH 5 ± 0.2 . The initial pH was adjusted by adding 0.01 M HNO_3 and/or NaOH as needed. The solution was agitated in a rotary shaker at 150 rpm. The supernatant was collected at different time intervals during 4-h of the experiment and analyzed for equilibrium metal concentration in the filtrate using AAS.

2.3. Metal elution from the preloaded biomass

The single and tri-metal laden biomasses were separately treated with HNO_3 (0.05 M) for 1 h on a rotary shaker at 100 rpm to elute the metal ions. After elution, the solutions were filtered and filtrate was measured for metal ion concentration using AAS. The biomass was regenerated by washing with deionized water till the pH of the wash solution reached 5, followed by shaking in a solution of Ca^{+2} and Mg^{+2} (0.01 M) for 30 min at 100 rpm (Xinjiang, 2006) so as to ensure more availability and homogeneity of the bioadsorbent sites. The regenerated biomass was air dried and reused up to two cycles to examine its capability for retaining metal adsorption.

3. Results and discussion

3.1. Effect of initial metal concentration

The maximum metal adsorption recorded from 40 mg l^{-1} of metal system (Fig. 1) could be ascribed to the fact that up to this metal concentration, a relatively high adsorptive surface area of biomass was available to the metal ions. But as the metal concentration increased

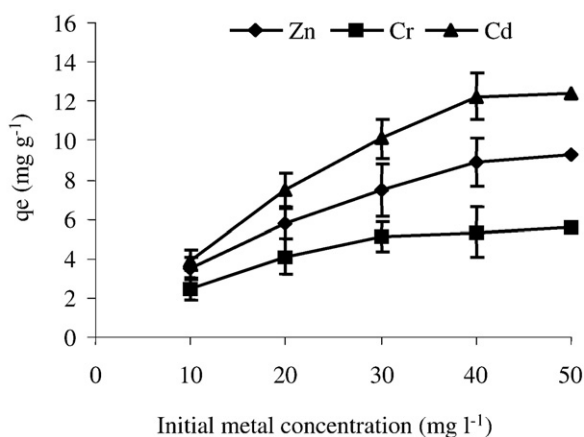


Fig. 1. Effect of concentration of Cd(II), Zn(II) and Cr(VI) metal ions on the metal adsorption capacity of *E. crassipes* dead biomass from single metal system. Vertical bars represent standard error values.

beyond 40 mg l^{-1} , the binding sites became saturated, as the amount of loaded biomass remained constant (Sekhar et al., 2003; Naja and Voelesky, 2006). Relatively greater ionic radii of Cd might have increased its affinity towards biomass. Tobin et al. (1984) and Al-Qunaibit (2009) postulated that larger metal ions are absorbed more strongly than smaller ones, primarily due to strong electrostatic interaction with binding sites of biomass.

3.2. Effect of pH on metal adsorption

Adsorption capacity for Cd and Zn was maximum at pH 5 and 6 respectively (Fig. 2), which could be attributed to less availability of H^+ to compete with these metals for adsorption sites of biomass (Patil et al., 2006; Shin et al., 2007). Earlier workers (Davis et al., 2003; Oh and Tshabalala, 2007; Verma et al., 2008) have suggested that carboxylic groups ($-\text{COOH}$) are the main adsorption sites, which deprotonate at pH range 3.5–5.5 and attract the positively charged metal ions to facilitate metal binding. Further, carboxylic groups having pK_a values about 4 indicated that most sites will be free and easily available for metal binding at $\text{pH} > 4$ (Schiewer, 1999), which matched well with the present study, especially for Zn and Cd. A relatively lower adsorption at pH below and above optimum values was due to heavy protonation of the negatively charged binding sites and formation of metal hydroxides respectively (Yan and Viraraghavan, 2003). However, optimum adsorption of Cr recorded at pH 2, showed conformity with the observations of Popuri et al. (2007) and Singh et al. (2008), who reported predominance of oxo-anions such as HCrO_4^{-1} , H_2CrO_4 and $\text{HCr}_2\text{O}_7^{-1}$ in strong acidic systems to accelerate metal adsorption.

3.3. Effect of biomass concentration and agitation time

The profiles of Zn and Cd adsorption capacity were very similar, showed a rapid increase initially and reached equilibrium within 120 min, whilst that of Cr took 180 min agitation to achieve equilibrium (Fig. 3). Such a behaviour could be ascribed to relatively smaller radii of Cr which took more agitation time to attain equilibrium. For each metal, the adsorption capacity increased with increasing biomass concentration (up to 0.5 g) leading to the corresponding increase in adsorption sites (Sekhar et al., 2003; Chubar et al., 2004) and stabilized afterward due to cell crowding and consequent reduction in intercellular distance (Parvathi et al., 2007) resulting in sub-optimal electrostatic interactions due to swathing of binding sites from metal ions (Pons and Fuste, 1993). In general, the maximum adsorption of Cd followed by Zn and Cr was recorded, which could be attributed to the difference in their ionic radii vis-a-vis ionic interaction with varying functional groups present on biomass surface (Tobin et al., 1984), however, this yet to be confirmed.

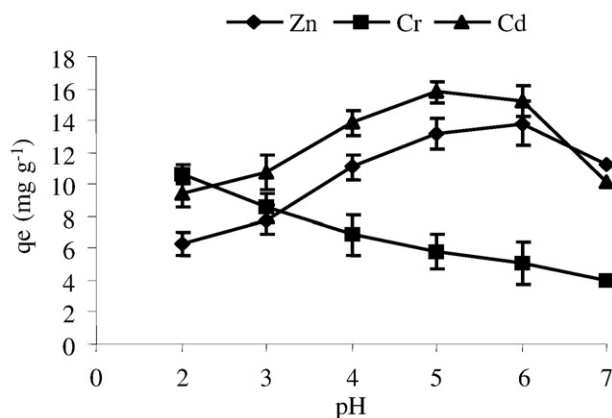


Fig. 2. Effect of pH on metal adsorption capacity of *E. crassipes* dead biomass from single metal system of Cd(II), Zn(II) and Cr(VI). Vertical bars represent standard error values.

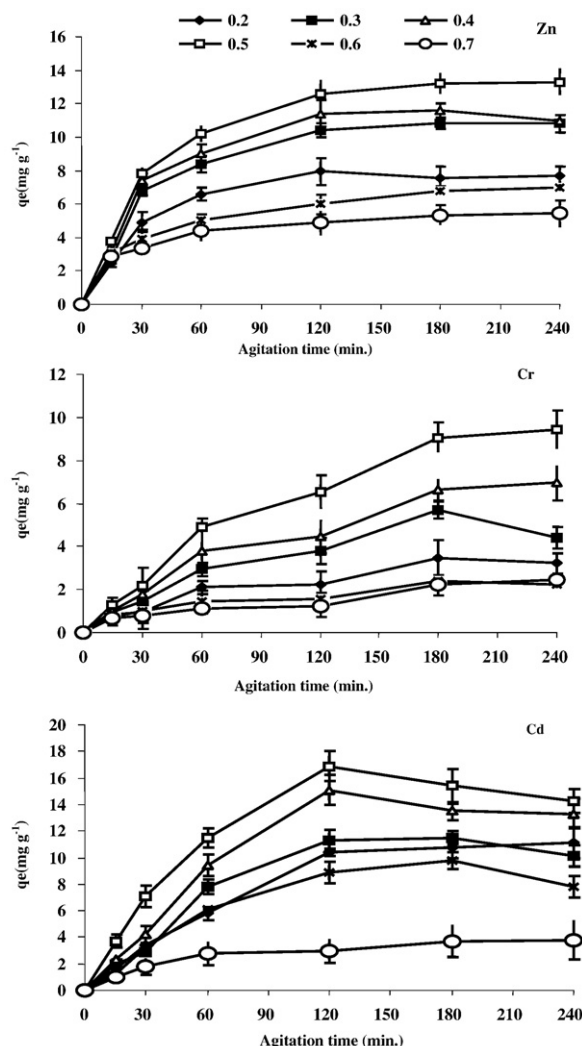


Fig. 3. Effect of biomass concentration (0.2 to 0.7 g) and agitation time on metal adsorption capacity of *E. crassipes* dead biomass from single metal system. Vertical bars represent standard error values.

3.4. Metal adsorption in bi- and tri-metal metal systems

In general, the metal adsorption decreased in both bi- and tri-metal systems at each concentration level except Cd, which showed an increase in bi-metal systems (Table 1). Higher Cd adsorption demonstrated its relatively higher affinity to binding sites/organic ligands owing to its greater ionic radii and conformed with the observations of Jang et al. (2005) and Miretzky et al. (2006). However, reduced metal adsorption in trimetal system as compared to that in a single metal system could be ascribed to crowding of all the metal ions in a constant volume (200 ml) solution and saturation of binding sites in constant biomass concentration (0.5 g), leading to enhanced magnitude of competition between metal ions for binding sites (Li et al., 2004; Hawari and Mulligan, 2006), wherein Cd superseded Zn and Cr. As such, the possible occurrence of complex electrostatic interaction between metal ions and multi-variables (pH, ionic strength or metal/site ratio) dependent acid–base properties of biomass (Lodeiro et al., 2007) leading to variation in its metal adsorption capacity cannot be ruled out, especially in bi- and tri-metal systems (Pal et al., 2009).

3.5. Metal elution and biomass regeneration

The metal recovery was higher in a single metal system than a trimetal system (Table 2) primarily due to less competition of the ions

Table 1

Adsorption (mg g^{-1}) of Zn(II), Cd(II) and Cr(VI) metal ions on dead biomass of *E. crassipes* in single, bi- and tri-metal systems ($\pm$ S.E.).

	Zn(II)	Cd(II)	Cr(VI)
Individual metal (20 mg l^{-1})	5.78 ± 0.46	7.45 ± 0.31	4.11 ± 0.27
Individual metal (40 mg l^{-1})	8.90 ± 1.07	12.26 ± 1.88	5.34 ± 0.62
Zn + Cd (20 mg l^{-1} each)	4.85 ± 0.62	14.18 ± 1.36	–
Zn + Cd (40 mg l^{-1} each)	4.04 ± 0.56	14.56 ± 1.12	–
Zn + Cr (20 mg l^{-1} each)	4.88 ± 0.37	–	3.51 ± 0.48
Zn + Cr (40 mg l^{-1} each)	3.76 ± 0.52	–	3.02 ± 0.15
Cr + Cd (20 mg l^{-1} each)	–	12.03 ± 0.28	3.46 ± 0.10
Cr + Cd (40 mg l^{-1} each)	–	7.07 ± 0.45	2.67 ± 0.52
Zn + Cd + Cr (20 mg l^{-1} each)	4.18 ± 0.21	6.87 ± 1.16	2.06 ± 0.08
Zn + Cd + Cr (40 mg l^{-1} each)	3.38 ± 0.13	4.56 ± 0.44	1.92 ± 0.17

for adsorption sites in the former. The maximum recovery was recorded for Cd (99%) followed by Zn (97%) and Cr (83%). The regenerated biomass exhibited increased metal adsorption up to two subsequent cycles and decreased afterwards. The increased metal adsorption may be due to the enrichment of biomass with Ca^{+2} and Mg^{+2} which plays an important role in the ion-exchange process by competing with other metal cations, especially Cd during metal adsorption (Gunneriusson and Sjöberg, 1991; Verma et al., 2008). Further, the decreased metal adsorption after the second regeneration cycle may be due to the loss of Ca^{+2} and Mg^{+2} ions from biomass as a result of adsorption–elution of metals by acid treatment (Xinjiao, 2006).

4. Conclusions

The study clearly showed that removal of Cd(II), Zn(II) and Cr(VI) from single, bi- and tri-metal systems by *E. crassipes* dead biomass through ion-exchange and electrostatic interaction leading to complexation with binding sites are principal mechanisms of metal adsorption, which depend on metal and biomass concentrations, pH and agitation time. The metal removal pattern followed $\text{Cd} > \text{Zn} > \text{Cr}$. Further, the recovery of adsorbed metal ions and the ability of biomass to retain its adsorption capacity up to two subsequent cycles demonstrated that *E. crassipes* dead biomass could be utilized for efficient adsorption and recovery of toxic metals from the industrial effluents, which invariably are multimetal systems.

Table 2

Adsorption (mg g^{-1}) and elution (mg l^{-1}) of Zn(II), Cd(II) and Cr(VI) metal ions with the dead biomass of *E. crassipes* from single and tri-metal systems up to two regeneration cycles ($\pm$ S.E.).

Metal ions		Single metal system (40 mg l^{-1})		Tri-metal system (40 mg l^{-1} each)	
		I cycle	II cycle	I cycle	II cycle
Zn (II)	Adsorbed	4.04 ± 0.53	3.63 ± 0.42	2.96 ± 0.35	2.53 ± 0.49
	Eluted	18.38 ± 2.15	17.60 ± 2.64	11.54 ± 2.02	11.26 ± 2.33
	Recovered (%)	91	97	78	89
Cd (II)	Adsorbed	4.67 ± 0.38	4.45 ± 0.26	4.19 ± 0.68	4.36 ± 0.42
	Eluted	22.18 ± 2.73	22.03 ± 1.82	18.02 ± 1.56	19.84 ± 2.31
	Recovered (%)	95	99	86	91
Cr (VI)	Adsorbed	2.65 ± 0.46	1.88 ± 0.11	2.37 ± 0.17	2.54 ± 0.22
	Eluted	11.09 ± 1.76	7.24 ± 1.41	7.94 ± 0.96	8.38 ± 1.79
	Recovered (%)	83	77	67	66

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